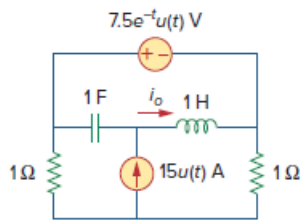


Problem

Calculate $i_o(t)$ for $t > 0$ in the network of Fig. 16.79.

Figure 16.79:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 16.79 in the textbook.

It is known that,

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

At $t = 0^-$, both current and voltage source values are zero. So, the initial current through the inductor and initial voltage across the capacitor are zero.

For $t > 0$ both sources are active.

The Laplace transform of $7.5e^{-t}u(t)$ is,

$$L[7.5e^{-t}u(t)] = \frac{7.5}{s+1}$$

Step 2 of 7

Draw the s-domain circuit diagram for $t > 0$

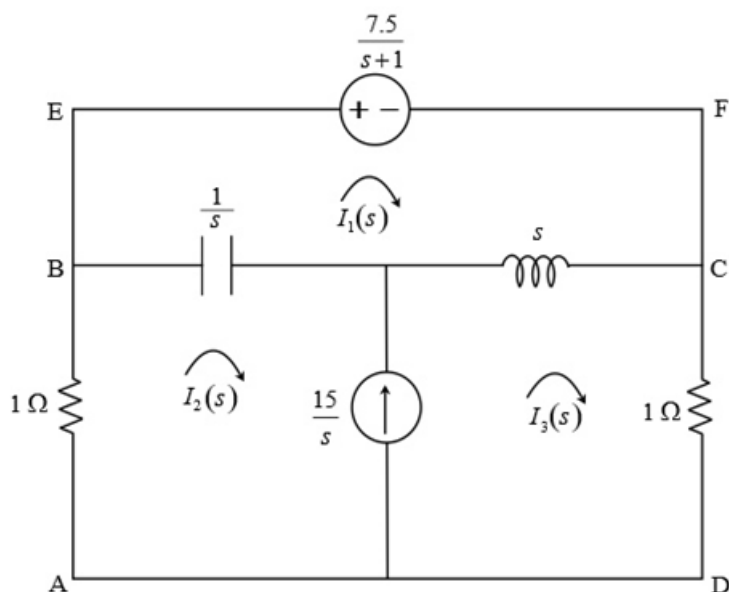


Figure 1

Step 3 of 7

A current source is between meshes 2 and 3, and it forms super mesh. The current source expression is,

$$I_3(s) - I_2(s) = \frac{15}{s}$$

$$I_2(s) = I_3(s) - \frac{15}{s} \dots\dots (1)$$

Apply Kirchhoff's voltage law to loop $I_1(s)$.

$$\frac{7.5}{s+1} + s[I_1(s) - I_3(s)] + \frac{I_1(s) - I_2(s)}{s} = 0$$

$$\left(s + \frac{1}{s}\right)I_1(s) - \frac{1}{s}I_2(s) - sI_3(s) = -\frac{7.5}{s+1}$$

$$\left(\frac{s^2+1}{s}\right)I_1(s) - \frac{1}{s}I_2(s) - sI_3(s) = -\frac{7.5}{s+1}$$

Substitute $I_3(s) - \frac{15}{s}$ for $I_2(s)$.

$$\left(\frac{s^2+1}{s}\right)I_1(s) - \frac{1}{s}\left(I_3(s) - \frac{15}{s}\right) - sI_3(s) = -\frac{7.5}{s+1}$$

$$\left(\frac{s^2+1}{s}\right)I_1(s) - \left(\frac{1}{s} + s\right)I_3(s) + \frac{15}{s^2} = -\frac{7.5}{s+1}$$

$$\left(\frac{s^2+1}{s}\right)I_1(s) - \left(\frac{1+s^2}{s}\right)I_3(s) = -\frac{7.5}{s+1} - \frac{15}{s^2}$$

$$I_1(s) - I_3(s) = -\left(\frac{7.5}{s+1} + \frac{15}{s^2}\right)\left(\frac{s}{s^2+1}\right) \dots\dots (2)$$

Step 4 of 7

Calculate the current $I_o(s)$.

$$I_o(s) = I_3(s) - I_1(s)$$

$$= -[I_1(s) - I_3(s)]$$

Substitute $-\left(\frac{7.5}{s+1} + \frac{15}{s^2}\right)\left(\frac{s}{s^2+1}\right)$ for $I_1(s) - I_3(s)$.

$$I_o(s) = -\left[-\left(\frac{7.5}{s+1} + \frac{15}{s^2}\right)\left(\frac{s}{s^2+1}\right)\right]$$

$$= \left(\frac{7.5s^2 + 15s + 15}{s^2(s+1)}\right)\left(\frac{s}{s^2+1}\right)$$

$$= \frac{7.5(s^2 + 2s + 2)}{s(s+1)(s^2+1)}$$

Step 5 of 7

Solve the equation using partial fractions.

$$I_o(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{Cs+D}{s^2+1} \dots\dots (3)$$

$$\frac{7.5(s^2+2s+2)}{s(s+1)(s^2+1)} = \frac{A(s+1)(s^2+1) + Bs(s^2+1) + (Cs+D)s(s+1)}{s(s+1)(s^2+1)}$$

$$7.5(s^2+2s+2) = A(s+1)(s^2+1) + Bs(s^2+1) + (Cs+D)s(s+1) \dots\dots (4)$$

Substitute 0 for s .

$$7.5(0^2+2(0)+2) = A(0+1)(0^2+1) + B(0)(0^2+1) + (C(0)+D)(0)(0+1)$$

$$15 = A$$

Substitute -1 for s in equation (4).

$$7.5[(-1)^2+2(-1)+2] = \left\{ A(-1+1)[(-1)^2+1] + B(-1)[(-1)^2+1] + \right. \\ \left. [C(-1)+D](-1)(-1+1) \right\}$$

$$7.5[1-2+2] = 0 + B(-1)[1+1] + [-C+D](-1)(0)$$

$$7.5 = -2B$$

$$B = -3.75$$

Step 6 of 7

Equate s coefficients form equation (4).

$$15 = A + B + D$$

$$15 = 15 - 3.75 + D$$

$$D = 3.75$$

Equate s^3 coefficients form equation (4).

$$0 = A + B + C$$

$$0 = 15 - 3.75 + C$$

$$C = -11.25$$

Step 7 of 7

Recall equation (3).

$$I_o(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{Cs+D}{s^2+1}$$

Substitute 15 for A , -3.75 for B , -11.25 for C and 3.75 for D .

$$I_o(s) = \frac{15}{s} + \frac{-3.75}{s+1} + \frac{11.25s+3.75}{s^2+1}$$

$$= \frac{15}{s} - \frac{3.75}{s+1} + \frac{11.25s}{s^2+1} + \frac{3.75}{s^2+1}$$

Apply inverse Laplace transform.

$$i_o(t) = 15u(t) - 3.75e^{-t}u(t) + 11.25\cos(t)u(t) + 3.75\sin(t)u(t)$$

$$= [15 - 3.75e^{-t} + 11.25\cos(t) + 3.75\sin(t)]u(t) \text{ A}$$

Therefore, the current $i_o(t)$ is $\boxed{[15 - 3.75e^{-t} + 11.25\cos(t) + 3.75\sin(t)]u(t) \text{ A}}$.